ACADEMIC LITERATURE DATABASE | PLANT BIOMONITORING PROJECT

# Scientific Review & Synthesis

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## Accumulation of particulate matter and trace elements on vegetation as affected by pollution level, rainfall and the passage of time

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### 1. Abstract & Scope

Air pollution is a major environmental threat. We evaluated PM and trace element accumulation on Pinus sylvestris (Scots pine) and Hedera helix (common ivy) over time, and the impact of heavy rainfall and pollution levels in Norway and Poland. Conifers with thick epicuticular wax (715.6 µg/cm² on pine) acted as permanent sinks that retained over 85% of PM against rainfall. Ivy (wax content 45.2 µg/cm²) experienced rapid cuticular saturation and lost up to 90% of its accumulated PM (SPM) during precipitation. Rainfall strips coarse surface particles but leaves the wax-embedded fraction unaffected.

### 2. Methodology & Experimental Parameters

Longitudinal sampling at industrial, traffic, and clean locations. Monitoring before and after natural rain events. Gravimetric filtration and acid digestion.

### 3. Core Findings & Data Synthesis

Pinus sylvestris accumulated up to 417.6 µg/cm² and ivy up to 140.6 µg/cm² of PM. Rain washed off 30-41% of coarse surface particles, but wax-embedded fine particles were shielded from hydraulic shear.

### 4. Strategic Relevance to JU and Kolkata Planning

Highly relevant to monsoon zones (such as West Bengal). Emphasizes that high-wax species are required to prevent rain from flushing toxic PM into municipal water drains.